EXERCISES 3.3

Answers to selected odd-numbered problems begin on page ANS-11.

In Problems 1 and 2, the given function is analytic for all z. Show that the Cauchy-Riemann equations are satisfied at every point.

1. $f(z) = z^3$

2. $f(z) = 3z^2 + 5z - 6i$

In Problems 3–8, show that the given function is not analytic at any point.

3. $f(z) = \text{Re}(z)$

4. $f(z) = y + ix$

5. $f(z) = 4z - 6\bar{z} + 3$

6. $f(z) = \bar{z}^2$

7. $f(z) = x^2 + y^2$

8. $f(z) = \dfrac{x}{x^2 + y^2} + i\dfrac{y}{x^2 + y^2}$

In Problems 9–16, (**a**) use Theorem 3.3.2 to show that the given function is analytic in an appropriate domain, and (**b**) use (9) or (11) to find the derivative of the function in the domain.

9. $f(z) = e^{-x}\cos y - ie^{-x}\sin y$

10. $f(z) = x + \sin x \cosh y + i(y + \cos x \sinh y)$

11. $f(z) = e^{x^2-y^2}\cos 2xy + ie^{x^2-y^2}\sin 2xy$

12. $f(z) = 4x^2 + 5x - 4y^2 + 9 + i(8xy + 5y - 1)$

13. $f(z) = \dfrac{x-1}{(x-1)^2 + y^2} - i\dfrac{y}{(x-1)^2 + y^2}$

14. $f(z) = \dfrac{x^3 + xy^2 + x}{x^2 + y^2} + i\dfrac{x^2y + y^3 - y}{x^2 + y^2}$

15. $f(z) = \dfrac{\cos\theta}{r} - i\dfrac{\sin\theta}{r}$

16. $f(z) = 5r\cos\theta + r^4\cos 4\theta + i(5r\sin\theta + r^4\sin 4\theta)$

In Problems 17–20, find real constants a, b, c, and d so that the given function is entire.

17. $f(z) = 3x - y + 5 + i(ax + by - 3)$

18. $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$

19. $f(z) = 1 + 5r\cos\theta + ar\sin\theta + i\,(3 + br\sin\theta + r\cos\theta)$

20. $f(z) = r^a\cos 2\theta + br\sin\theta + i\left(cr^2\sin\theta\cos\theta + r\cos\theta\right)$

In Problems 21–24, (**a**) show that the given function is not analytic at any point but is differentiable along the indicated curve(s), and (**b**) use (9) or (11) to find the derivative of the function on the curve(s).

21. $f(z) = x^2 + y^2 + 2ixy$; x-axis

22. $f(z) = 3x^2y^2 - 6ix^2y^2$; coordinate axes

23. $f(z) = x^3 + 3xy^2 - x + i(y^3 + 3x^2y - y)$; coordinate axes

24. $f(z) = x^2 - x + y + i(y^2 - 5y - x)$; $y = x + 2$

25. In Section 2.1 we defined the complex exponential function $f(z) = e^z$ in the following manner $e^z = e^x\cos y + ie^x\sin y$.

(**a**) Show that $f(z) = e^z$ is an entire function.

(**b**) Show that $f'(z) = f(z)$

26. In Section 3.1 we defined the principal branch of the complex square root in the following manner $z^{1/2} = \sqrt{r}e^{i\theta/2}$, $-\pi < \theta < \pi$.

(**a**) Use (10) to show that $f_1(z) = z^{1/2}$ is analytic in the domain $-\pi < \theta < \pi$.

(**b**) Use (11) to show that $f_1'(z) = \frac{1}{2}z^{-1/2}$ in this domain.

Focus on Concepts

27. Suppose $f(z)$ is analytic. Show that $|f'(z)|^2 = u_x^2 + v_x^2 = u_y^2 + v_y^2$.

28. Suppose $u(x, y)$ and $v(x, y)$ are the real and imaginary parts of an analytic function f. Can $g(z) = v(x, y) + iu(x, y)$ be an analytic function? Discuss and defend your answer with sound mathematics.

29. Suppose $f(z)$ is entire. Use the Cauchy-Riemann equations to show that the function $g(z) = \overline{f(\overline{z})}$ is entire.

30. If $f(z)$ and $\overline{f(z)}$ are both analytic in a domain D, then what can be said about f throughout D?

31. In this problem you are guided through the start of the proof of the proposition:

> *If f is analytic in a domain D, and $|f(z)| = c$, where c is constant, then f is constant throughout D.*

Proof We begin with the hypothesis that $|f(z)| = c$. If $f(z) = u(x, y) + iv(x, y)$ then $|f(z)|^2 = c^2$ is the same as $u^2 + v^2 = c^2$. The partial derivatives of the last expression with respect to x and y are, respectively,

$$2uu_x + 2vv_x = 0 \quad \text{and} \quad 2uu_y + 2vv_y = 0.$$

Complete the proof by using the Cauchy-Riemann equations to replace v_x and v_y in the last pair of equations. Then solve for u_x and u_y and draw a conclusion. Use the Cauchy-Riemann equations again and solve for v_x and v_y.

32. If $f(z) = \overline{z}/z$, then show that $|f(z)|$ is constant in the domain defined by $z \neq 0$. Explain why this does not contradict Theorem 3.3.3(*i*).

33. Suppose $g(x)$ is a twice differentiable real function and f is a complex function defined by $f(z) = g(x) + iy^2$. Determine where $f(z)$ is differentiable and where it is analytic. Where did you use the hypothesis that g is twice differentiable?

34. Find a nonconstant complex function f that is nowhere analytic but is differentiable on the unit circle $|z| = 1$. [*Hint*: Consider using a function f similar to the one in Problem 33.]

35. In this problem you are guided through the start of the proof of the proposition:

> *If f is analytic in a domain D, and $f'(z) = 0$, then f is constant throughout D.*

Proof We begin with the hypothesis that f is analytic in D and hence it is differentiable throughout D. Hence by (9) of this section and the assumption that $f'(z) = 0$ in D, we have $f'(z) = \dfrac{\partial u}{\partial x} + i\dfrac{\partial v}{\partial x} = 0$. Now complete the proof.

36. Use the Cauchy-Riemann equations to prove that if f is analytic in a domain D, and if $f'(z)$ is a constant, then $f(z)$ must be a complex linear function throughout D.

37. Use the proposition in Problem 35 to show that if f and g are analytic and $f'(z) = g'(z)$, then $f(z) = g(z) + c$, where c is a constant. [*Hint*: Form $h(z) = f(z) - g(z)$.]

38. Suppose $x = r\cos\theta$, $y = r\sin\theta$, and $f(z) = u(x, y) + iv(x, y)$. Show that

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta, \qquad \frac{\partial u}{\partial \theta} = -\frac{\partial u}{\partial x}r\sin\theta + \frac{\partial u}{\partial y}r\cos\theta \tag{12}$$

and

$$\frac{\partial v}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta, \qquad \frac{\partial v}{\partial \theta} = -\frac{\partial v}{\partial x}r\sin\theta + \frac{\partial v}{\partial y}r\cos\theta. \tag{13}$$

Now use (1) in the foregoing expressions for v_r and v_θ. By comparing your results with the expressions for u_r and u_θ, deduce the Cauchy-Riemann equations in polar coordinates given in (10).

39. Suppose the function $f(z) = u(r, \theta) + iv(r, \theta)$ is differentiable at a point z whose polar coordinates are (r, θ). Solve the two equations in (12) for u_x and then solve the two equations in (13) for v_x. Then show that the derivative of f at (r, θ) is

$$f'(z) = (\cos\theta - i\sin\theta)\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) = e^{-i\theta}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right).$$

40. Consider the function

$$f(z) = \begin{cases} 0, & z = 0 \\ \dfrac{z^5}{|z^4|}, & z \neq 0. \end{cases}$$

(**a**) Express f in the form $f(z) = u(x, y) + iv(x, y)$,

(**b**) Show that f is not differentiable at the origin.

(**c**) Show that the Cauchy-Riemann equations are satisfied at the origin. [*Hint*: Use the limit definitions of the partial derivatives $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, and $\partial v/\partial y$ at $(0, 0)$.]

3.4 Harmonic Functions

INTRODUCTION In Section 5.5 we shall see that when a complex function $f(z) = u(x, y) + iv(x, y)$ is analytic at a point z, then *all* the derivatives of f: $f'(z)$, $f''(z)$, $f'''(z)$, ... are also analytic at z. As a consequence of this remarkable fact, we can conclude that *all* partial derivatives of the real functions $u(x, y)$ and $v(x, y)$ are continuous at z. From the continuity of the partial derivatives we then know that the second-order mixed partial derivatives are equal. This last fact, coupled with the Cauchy-Riemann equations, will be used in this section to demonstrate that there is a connection between the real and imaginary parts of an analytic function $f(z) = u(x, y) + iv(x, y)$ and the second-order partial differential equation

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0. \tag{1}$$

This equation, one of the most famous in applied mathematics, is known as **Laplace's equation** in two variables. The sum $\dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2}$ of the two second-order partial derivatives in (1) is denoted by $\nabla^2\phi$ and is called the **Laplacian** of ϕ. Laplace's equation (1) is then abbreviated as $\nabla^2\phi = 0$.

Harmonic Functions

A solution $\phi(x, y)$ of Laplace's equation (1) in a domain D of the plane is given a special name.

DEFINITION 3.4.1 Harmonic Functions

A real-valued function ϕ of two real variables x and y that has continuous first and second-order partial derivatives in a domain D and satisfies Laplace's equation is said to be **harmonic** in D.

Harmonic functions are encountered in the study of temperatures and potentials. These applications are introduced in Section 3.5. In the next theorem we see that both the real and the imaginary parts of an analytic function are harmonic.